Alwin Joshy

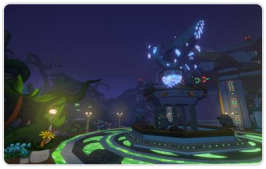
Technical Artist

Shaders · Rendering optimisation · Unity tools · Unity & Unreal

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SUMMARY

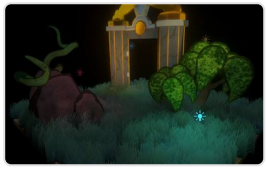
Technical Artist and Unity Engineer with 6+ years shipping real-time 3D on PC, VR, WebGL and mobile. I write shaders (HLSL, Shader Graph), build C# tools and pipelines, and own rendering performance. At Highstreet I led technical art for a VR / PC MMO and built imposter and voxel-lighting systems to keep dense scenes in budget. Today I am founder and technical artist at NPsynC, creating real-time digital humans in Unreal Engine.



Highstreet: VR / PC MMO (Unity)



Crystal shader (Shader Graph)



Voxel light baker (VR)



NPsynC: digital humans (Unreal)



HoloFair: WebGL virtual events

SKILLS

Technical art: Shader development (HLSL, Shader Graph, GLSL), VFX, C# editor tools and asset pipelines, technical art direction, bridging art and engineering

Rendering & optimisation: CPU/GPU profiling, draw-call reduction, SRP Batcher, GPU instancing, LOD and imposters, overdraw, memory management, compute shaders

Lighting: Real-time and baked lighting, light probes, reflections, post-processing, 3D-texture (voxel) light baking

Engines & platforms: Unity (expert, URP), Unreal Engine (MetaHuman, grooms), Three.js / WebGL; PC, VR, WebGL, mobile

Languages & tools: C#, C++, HLSL, JavaScript, Git, Blender, network programming, UI/UX programming

EXPERIENCE

Founder & Technical Artist | NPsynC

Feb 2026 – Present · Dubai, UAE

- Co-founded and run NPsynC, which provides AI digital-human staff (front-of-house, assistants, concierges) to government and semi-government agencies and companies.
- Create the company's digital humans in Unreal Engine (MetaHuman): modelling and texturing, rigging outfits, beard and eyebrow grooms, lighting and rendering.
- Characters talk in real time with live lipsync; NPsynC's own neural lipsync runs locally on the GPU and streams facial animation into Unreal Engine at 60 fps.

Unity Engineer & Technical Artist | Highstreet

Jan 2022 – Mar 2026 · Los Angeles, USA

- Led technical art and network programming for Highstreet's VR / PC MMO in Unity, and owned its performance and rendering budgets.
- Built a voxel light baker that bakes scene lighting into small 3D textures (one texture read per pixel, whatever the light count); a 45×34×45 volume is 268.9 KB, so each streamed VR scene block carries its own lighting. Used in the Arena and Lobby levels.
- When dense forest scenes dropped from 60–72 fps to 35–40 fps (sometimes ~15), built a circular-view imposter system: capture tool, texture atlas and a view-blending shader with normal / albedo relighting.
- Wrote custom shaders and tools for the environment art workflow, bridging the asset pipeline and the engine.
- Built VR grass: an editor painter, one combined mesh, proximity-activated zones and alpha clip with 4x MSAA, lit by the voxel light baker.
- Built a Shader Graph crystal shader: inset-edge meshes, parallax inner layers, sparkles, fake reflections and a growth dissolve.

Technical Artist & Network Programmer | **Knotgames**
Feb 2021 – Feb 2022

- Wrote custom HLSL shaders for specific art directions while holding 60 fps targets.
- Built network solutions and UI/UX systems for multi-user experiences, and coordinated the work.
- Owned QC, troubleshooting and optimisation of real-time rendering.

Technical Artist (Freelance) | **BlueMonocle Software**
Jun 2021 – Sep 2021

- Technical art direction for mobile and web projects.
- Earlier contract (Aug 2020 – Mar 2021): React Native front-end development.

C++ Student Support Tutor | **SAE Institute**
Jun 2020 – Aug 2021

- Tutored students in C++ memory management, pointers and data structures; post-lecture assistant.

Technical Artist & R&D (Internship) | **Outreal XR, HoloFair**
Nov 2019 – Aug 2020

- Built features in C# / Unity for HoloFair, an MMO-style virtual event platform in Unity WebGL and one of the first metaverse / virtual-event projects in the MENA region.
- Optimised assets and rendering so large shared event spaces ran smoothly in the browser.

Also: TrueBays Software, full-stack React Native developer (freelance, Jan 2021 – Oct 2021); SAE Institute, game developer and designer (paid internship, Jun 2018 – Jun 2019).

PROJECTS

- **Portfolio hero scene** (Three.js / WebGL, 2026): Draco cut the land model from 17.65 MB to 1.87 MB (3D payload ~12 MB to ~4.4 MB); clouds in one instanced draw call; adaptive quality below 45 fps; fog matched to the sky.
- **Time of Day Lighting Study** (Unity URP, 2026): One cabin scene lit for dawn, noon, dusk and night from the sun or moon, sky ambient, controlled bounce, reflections and a few support lights.
- **Dreamtopia** (Steam, 2020 · Lead Developer): Took the game from prototype to Steam release with a team at SAE.
- **Pablo Let's Go Bar** (Steam, 2021 · Game Programmer & Rendering): Gameplay and rendering programming; won Best Project at the SAE Showcase 2021.
- **BlackSky** (Steam · Shader Artist (contributor)): Contributed shader work.
- **Mesh to SDF in Unity** (Ray marching, 2026): Converts meshes to signed distance fields stored in small 3D textures and ray-marches them.

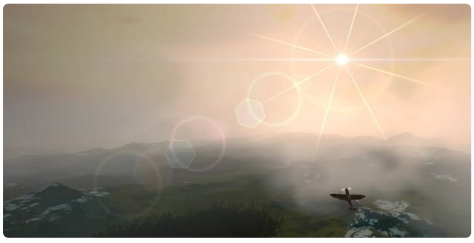
EDUCATION

Bachelor of Games Development (Games Programming) | **SAE Institute**
2018 – 2021 · GPA 6.5 / 7.0

ACHIEVEMENTS

- SAE Showcase Award 2021, Best Project: Pablo Let's Go Bar (Jun 2021).
- Shipped Dreamtopia on Steam (Jan 2020).
- Certificate of Appreciation, Liv. Hackathon, Amity University, Dubai Entrepreneurship Week (Oct 2019).

Languages: English (fluent), Spanish (conversational)



See the work in motion

Videos, shader breakdowns and write-ups for every project above.

alwinjoshy.github.io